

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T16 · Algebra Toolkit

With Answers

Questions with full worked solutions

■ 13 pages

FAMILY MAP

Algebra Toolkit

F01

Change a subject that appears twice or as a power

5 Q | 19 marks

F02

Substitute into an expression or formula

1 Q | 3 marks

Question 20

4MA1-2H Q15

ORIGINAL QUESTION

4 marks

15 (a) Make g the subject of $e = \sqrt{\frac{7g+5}{11+2g}}$

(4)

Worked Solution 20

Change the subject

MODULAR UNIT 2

4 marks

Square and remove the fraction

$$e^2(11 + 2g) = 7g + 5.$$

Collect the terms in the new subject

$$g(2e^2 - 7) = 5 - 11e^2 \Rightarrow g = \frac{5 - 11e^2}{2e^2 - 7}.$$

Final answer

$$g = \frac{5 - 11e^2}{2e^2 - 7}$$

Question 23

4MA1-2H Q20

ORIGINAL QUESTION

3 marks

20 Given that $k = x - y$ and $x = \frac{1}{4y}$

express $\frac{5k}{x+2}$ in the form $\frac{a-by^2}{c+dy}$ where a, b, c and d are integers.

Worked Solution 23

Algebraic substitution

MODULAR UNIT 2

3 marks

Substitute

$$x = 1/(4y), \quad k = 1/(4y) - y.$$

Simplify

$$\frac{5k}{x+2} = \frac{5-20y^2}{1+8y}.$$

Final answer

$$\frac{5-20y^2}{1+8y}$$

Question 12

4MA1-1H Q15 demand

ORIGINAL QUESTION UNIT 2

3 marks

15 Make n the subject of the formula $x = \frac{3p+n}{3n-4}$

.....

Worked solution 12Make n the subject

MODULAR UNIT 2

3 marks

Demand – Make n the subject

Clear the denominator

$$x(3n - 4) = 3p + n \Rightarrow 3nx - 4x = 3p + n.$$

Collect n terms

$$n(3x - 1) = 3p + 4x.$$

Divide

$$n = \frac{3p + 4x}{3x - 1}.$$

Final answer

$$n = (3p + 4x)/(3x - 1)$$

Question 27

4MA1-2H Q17

ORIGINAL QUESTION

4 marks

(b) Make e the subject of $w = \sqrt{\frac{e+g}{ef-d}}$

.....
(4)

Worked Solution 27Make e the subject of a square-root formula

MODULAR UNIT 2

4 marks

Square

$$w^2 = \frac{e + g}{ef - d}.$$

Clear the denominator

$$w^2(ef - d) = e + g \Rightarrow w^2ef - w^2d = e + g.$$

Collect e

$$e(w^2f - 1) = g + w^2d.$$

Divide

$$e = \frac{g + w^2d}{w^2f - 1}.$$

Final answer

$$e = \frac{g + w^2d}{w^2f - 1}$$

Original question 26

4MA1-2H Q12 M02

MODULAR UNIT 2

4 marks

(b) Make c the subject of the formula $f = \sqrt{\frac{a+bc}{c-d}}$

.....
(4)

Worked solution 26Make c the subject

MODULAR UNIT 2

4 marks

M02 – Make c the subject**Square**

$$f^2 = (a + bc)/(c - d).$$

Multiply out

$$cf^2 - df^2 = a + bc.$$

Collect c

$$c(f^2 - b) = a + df^2.$$

Divide

$$c = (a + df^2)/(f^2 - b).$$

Final answer

$$c = \frac{a + df^2}{f^2 - b}$$

Original question 27

4MA1-2H Q14 Q14(b)

ORIGINAL QUESTION

4 marks

(b) Make c the subject of $g = \frac{c+3}{4+c} - 7$

(4)

Worked solution 27

Chapter: Algebra Toolkit

MODULAR UNIT 2

4 marks

Q14(b) – Chapter: Algebra Toolkit

Family: Change a subject that appears twice or as a power • Variant: Subject appears in two linear terms

Method

$$g + 7 = (c + 3)/(c + 4).$$

*Why: Add 7 to both sides.***Method**

$$(g + 7)(c + 4) = c + 3, \text{ so } gc + 7c + 4g + 28 = c + 3.$$

*Why: Remove the fraction and expand.***Method**

$$c(g + 6) = -(4g + 25).$$

*Why: Collect all c-terms on one side.***Method**

$$\text{Therefore } c = -(4g + 25)/(g + 6).$$

*Why: Divide by the coefficient of c.***Final answer**

$$\text{c: } c = -(4g + 25)/(g + 6)$$